

CAVEAT: A Multi-View Benchmark for Human Action Recognition in Immersive CAVE Environments

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Abstract

Human action recognition (HAR) inside large-scale immersive systems such as CAVEs (Cave Automatic Virtual Environments) promises marker-less interaction, automatic procedure assessment, and behavioral analysis for safety-critical training. Yet every widely used HAR benchmark implicitly assumes conditions that a CAVE violates: frontal tripod-height viewpoints, stable photometric backgrounds, and offline evaluation. We present CAVEAT, the first HAR benchmark captured inside an operational projection-based CAVE. It provides synchronized multi-view RGB video from seven cameras—including the top-mounted, steeply pitched views forced by CAVE geometry—together with optical-motion-capture-grade 3D skeleton ground truth (SMPL layout, 22 joints) for 15 actors performing 7 general and 11 VR-interaction actions across 6 projected scenes. Because the projected VR content changes the entire visual background under controlled conditions, CAVEAT additionally enables a background-shift evaluation that is impossible to stage in real-world scenes. We define four evaluation protocols (cross-subject, cross-view, cross-scene, and cross-domain from NTU RGB+D 120), plus a latency-aware protocol that reports computation and observation latency for streaming recognition. Benchmarking [TODO: N (G13)] representative skeleton, video, and multimodal models reveals [TODO: 核心发现一句话, 需实验后填 (G13)]. The dataset, evaluation code, and correction toolchain will be publicly released.

1 Introduction

Projection-based immersive systems such as the CAVE [1] remain the platform of choice when virtual reality must be combined with free full-body movement, unobstructed vision of one’s own body, and multi-user co-presence—requirements typical of industrial safety training, rehabilitation, and simula-

tion. Human action recognition inside such environments would enable automatic evaluation of trainee procedures, marker-less interaction, and quantitative behavioral analysis.

Modern HAR models are, however, trained and evaluated almost exclusively on datasets that encode three assumptions. First, *fixed frontal geometry*: cameras in NTU RGB+D [2, 3] and its successors observe subjects from tripod height at moderate distances, whereas a CAVE forces cameras to its ceiling rim, producing steep pitch angles and systematic self-occlusion of hands and head during interaction. Second, *appearance stability*: laboratory backgrounds are static, whereas a CAVE’s walls and floor *are* the display—the background changes radically with the projected scene and its light spills onto the actor. Third, *offline evaluation*: accuracy is reported on trimmed clips without latency constraints, whereas interactive VR imposes tight motion-to-response budgets. The gap between these assumptions and CAVE reality is unquantified because no suitable benchmark exists.

We close this gap with CAVEAT¹, captured in a [TODO: CAVE 尺寸/面数确认: 幻灯片同时出现 ``5 faces active" 与 ``6-sided" (G2)] projection CAVE instrumented with 12 OptiTrack Prime^x13 motion-capture cameras and 7 RGB cameras. Our contributions are:

- **Dataset.** [TODO: 清洗后片段总数 (G8)] clips of 15 actors (9M/6F), 6 projected scenes (3 indoor, 3 outdoor), 7 general + 11 VR-interaction actions, with synchronized 7-view RGB and mocap-grade SMPL-22 skeletons—to our knowledge the first HAR corpus recorded inside an operational CAVE, and the first with top-mounted viewpoints as a primary condition.
- **Human-verified ground truth.** A correction pipeline combining per-clip quality scor-

¹CAVEAT = **CAVE** **AcTions** — also, we hope, a literal *caveat* for action recognition models benchmarked only in conventional laboratories.

ing, two-view triangulated editing, and bone-length-preserving inverse-kinematics tools; all released test-split skeletons are manually verified ([TODO: 修正帧占比 X% (G9)], inter-annotator agreement [TODO: Y mm (G10)]).

- **Protocols.** Cross-subject, cross-view, cross-scene, and NTU→CAVEAT cross-domain splits, plus a latency-aware streaming protocol reporting computation and observation latency.
- **Baselines.** [TODO: N (G13)] models spanning skeleton GCNs and transformers, video networks, and multimodal fusion, with an analysis of where and why they fail under CAVE conditions.

2 Related Work

HAR datasets. Existing corpora split into two families with an inherent scale-fidelity trade-off. The *scale* family—NTU RGB+D [2], NTU RGB+D 120 [3], ANUBIS [4], Kinetics [5]—recruits large cohorts but estimates skeletons from consumer depth sensors or 2D pose, with joint errors on the order of centimeters and frequent tracking failure under occlusion. The *fidelity* family anchors joints with optical motion capture and has always traded cohort size for precision: Human3.6M [6] records 11 actors with 4 cameras, TotalCapture [7] 5 actors with 8 cameras. CAVEAT belongs to the fidelity family—240 Hz optical mocap refined by MoSh++ and manual verification—while providing the largest actor cohort in that family, the most action classes, and conditions none of them cover: top-mounted viewpoints, prop-based VR interaction, and controlled background shift (Table 2).

User behavior corpora in VR. LocoVR [8] records multi-user locomotion trajectories in HMD-based VR; HUMOTO [9] provides mocap human-object interactions without multi-view RGB. HMD-based corpora observe users wearing headsets that occlude the face and alter motion; CAVE users wear only lightweight shutter glasses, keeping appearance close to natural while introducing the projection-specific challenges we target. RoCoG-v2 [10] studies synthetic-to-real domain shift for gestures, methodologically related to our cross-domain protocol.

Models. Skeleton-based recognition evolved from graph convolutions (ST-GCN [11], CTR-GCN [12]) to attention and prototype-based designs (SkateFormer [13], ProtoGCN [14]), state-space models

(SkelMamba [15]), and pretrained motion encoders (MotionBERT [16]). Video models progressed from two-pathway CNNs (SlowFast [17]) to transformers (Video Swin [18], VideoMAE v2 [19]) and state-space architectures (VideoMamba [20]); language-vision pretraining enables zero-shot transfer (ActionCLIP [21]); and PoseConv3D [22] fuses pose heatmaps with RGB. We benchmark representatives of each family.

Online and latency-aware evaluation. Online action detection [23] and streaming perception argue that offline accuracy overstates deployable performance; recent work separates *computation* latency from *observation* latency [24]. VR interaction makes these constraints first-class, motivating our protocol.

3 The CAVEAT Dataset

3.1 Capture environment

Recording took place in the [TODO: 机构匿名化处理后的 CAVE 描述 + 尺寸 (G2)] CAVE. Twelve OptiTrack Prime^x13 cameras capture 3D motion at 240 Hz; seven RGB cameras ([TODO: 型号/分辨率/帧率 (G3)], nominally 30 fps) cover the volume in a 3×2 grid on the left, center, and right rim at high and low mounting positions, plus one diagonal rear view. Fig. ??[TODO: 相机布置图 (G4)] shows the layout; the “high” row is top-mounted at pitch angles of [TODO: 俯仰角度数 (G3)], the geometry that distinguishes CAVE deployments from tripod-based datasets.

Two collection rounds were performed. Round 1 (portal-configuration LED CAVE, [TODO: 7 或 8 (幻灯片不一致, G5)] mocap + 2 RGB cameras) served as a pilot; Round 2, described above, is the release configuration. Actors wore everyday clothing with 37 retro-reflective markers; Round 2 used low-profile marker stickers to minimize visual interference with the RGB streams.

3.2 Actors, scenes, and actions

Fifteen actors (9 male, 6 female; [TODO: 年龄范围/身高范围, 需 roster (G6)]) each performed in six projected scenes (three indoor, three outdoor), approximately 20 effective minutes per scene and roughly 41,400 mocap frames per actor. The action vocabulary (Table 1) contains seven general actions shared with existing benchmarks—enabling direct cross-domain comparison—and eleven VR-interaction actions elicited by the projected applica-

Table 1: Action vocabulary: 7 general + 11 VR-interaction classes.

General	VR-interaction
walking	move via controller
running	catch fish with net
jumping	grab box
bending down	measure length
standing	wave
squatting	throw
raising hand	cut
	shoot
	pick up item
	go up stairs
	dodge car

tions, from controller locomotion to catching virtual fish with a physical net prop.

3.3 Calibration and synchronization

RGB intrinsics and extrinsics follow Zhang’s method [25] with per-session refinement; extrinsic drift is monitored by re-projecting triangulated 2D poses and gating on median residual ([**TODO: 标定精度: 平均重投影误差 px (G11)**]). Lacking hardware genlock, video–skeleton alignment uses a per-clip timeline established from the mocap clock, refined per camera with sub-frame offsets and verified by overlay inspection; final alignment accuracy is [**TODO: 同步精度 $\pm$ 帧 (G11)**].

3.4 Ground-truth construction

Raw 240Hz markers are solved to SMPL [26] bodies with MoSh++ [27]; we release the 22-joint SMPL subset (hand joints excluded owing to sparse hand markers) in both world coordinates and pelvis-centered normalized form. Sparse marker coverage and interaction props cause characteristic failures—frozen or drifting limb segments—which we repair with a purpose-built toolchain: (i) every clip receives an automatic quality score combining bone-length deviation, robust velocity-spike detection, frozen-joint-while-moving runs, and missing-marker fraction, which prioritizes review worst-first; (ii) annotators correct joints by dragging in two calibrated views simultaneously (exact triangulated depth) with keyframe–spline propagation; and (iii) a bone-length-preserving two-bone IK mode solves whole-limb placements from a single wrist/ankle drag, guaranteeing physically plausible corrections that never fold a straight limb. All test-split clips are fully verified; training-split clips

are corrected worst-first by quality score with per-frame edit masks released ([**TODO: 银标准修正比例 (G9)**]). Corrections modified [**TODO: X% (G9)**] of test-split frames; dual-annotation of a 5% sample shows mean joint disagreement of [**TODO: Y mm (G10)**]. Fig. [**TODO: 标注工具界面图: 双视角+极线+IK黄圈, 一图讲清方法论 (G22)**] illustrates the tool; the full correction SOP, marker layout, and capture protocol appear in the supplementary material, and the toolchain itself is released so that others can extend the corpus at the same quality bar.

3.5 Quantifying the CAVE gap

To turn the three violated assumptions of §1 into measurements, we characterize the corpus along three axes; all quantities are computed directly from the released calibration and ground truth.

Viewpoint geometry. From the camera extrinsics we report the distribution of camera pitch angles relative to the subject’s pelvis across all frames: the top-mounted row observes actors at [**TODO: 中位俯仰角/范围, 由外参+骨架直接算 (G19)**], versus the near-horizontal geometry of tripod-based corpora.

Interaction-driven self-occlusion. Projecting the mocap ground truth into each view and testing body self-occlusion [**TODO: 遮挡判定方法一句话 (G20)**], wrists are occluded in [**TODO: X% (G20)**] of interaction-action frames in top-mounted views versus [**TODO: Y% (G20)**] in the low-rim views—occlusion statistics that ground-truth-free datasets cannot even measure.

Projected-background variation. Because the CAVE walls are displays, scene changes shift the entire background and spill colored light onto the actor. We quantify this as the inter-scene distance of background color statistics and the intra-actor appearance drift across scenes ([**TODO: LAB 直方图距离等具体度量与数字 (G21)**]), an order of magnitude [**TODO: 核实后措辞 (G21)**] beyond the illumination variation of laboratory corpora.

3.6 Statistics

[**TODO: 每类样本数/时长分布/marker 分布直方图, 清洗后生成 (G8)**]

最终统计表:
丢失率/QC

Table 2: Comparison with representative HAR and motion-capture datasets. Estimated-skeleton corpora (top) scale in subjects but provide noisy joints; mocap-grade corpora (bottom) trade cohort size for precision. CAVEAT offers the largest cohort and action vocabulary in the mocap-grade family, with capture conditions unavailable elsewhere. **[TODO: 逐格终核数字 (G7)]**

Dataset	Subjects	Classes	Skeleton GT (rate)	Calib. RGB views	Top-mounted	Props/VR
<i>Estimated skeletons (scale family)</i>						
NTU RGB+D 120 [3]	106	120	Kinect v2 (30 Hz)	3 viewpoints	X	X
ANUBIS [4]	80	102	Azure Kinect (30 Hz)	X	X	X
<i>Mocap-grade skeletons (fidelity family)</i>						
Human3.6M [6]	11	17 scen.	optical mocap (50 Hz)	4	X	X
TotalCapture [7]	5	5 acts	Vicon+IMU (60 Hz)	8	X	X
CAVEAT (ours)	15	18	optical mocap (240 Hz) + MoSh++	7		

4 Benchmark Protocols

Cross-subject (X-Sub, primary). Three fixed 10-train/5-test actor partitions (seeded, gender-balanced **[TODO: 性别核对后确认 (G12)]**); we report mean $\pm$ std over the three partitions to control for small-cohort variance.

Cross-view (X-View). Train on low-rim views, test on top-mounted and diagonal-rear views, isolating the viewpoint gap.

Cross-scene (X-Scene). Train on the three indoor projections, test on the three outdoor ones: identical actions, radically different projected backgrounds—a controlled background-shift experiment unavailable in real-world corpora.

Cross-domain. Models pretrained on NTU RGB+D 120 are evaluated zero-shot and after fine-tuning, quantifying the lab-to-CAVE domain gap that motivates this work.

Metrics. Top-1/Top-5 accuracy and mean per-class accuracy (the 18-class vocabulary is imbalanced across interaction types); confusion analyses separate general from interaction classes.

Latency-aware protocol. For streaming deployment we report (i) *computation latency*: single-clip inference time on one **[TODO: 评测 GPU 型号 (G14)]**, and (ii) *observation latency*: the minimum observed-frame budget at which a model reaches 90% of its full-clip accuracy [23, 24], yielding accuracy–latency curves (Fig. **[TODO: latency 图 (G15)]**).

Table 3: Baselines on CAVEAT (Top-1 / mean per-class, %). **[TODO: 全部数字 (G13)]**

Model	Modality	X-Sub	X-View	X-Scene
ST-GCN [11]	Skel.	[TODO:]	[TODO:]	[TODO:]
CTR-GCN [12]	Skel.	[TODO:]	[TODO:]	[TODO:]
SkateFormer [13]	Skel.	[TODO:]	[TODO:]	[TODO:]
SkelMamba [15]	Skel.	[TODO:]	[TODO:]	[TODO:]
MotionBERT [16]	Skel.	[TODO:]	[TODO:]	[TODO:]
SlowFast [17]	RGB	[TODO:]	[TODO:]	[TODO:]
Video Swin [18]	RGB	[TODO:]	[TODO:]	[TODO:]
VideoMAE v2 [19]	RGB	[TODO:]	[TODO:]	[TODO:]
VideoMamba [20]	RGB	[TODO:]	[TODO:]	[TODO:]
ActionCLIP [21]	RGB+T	[TODO:]	[TODO:]	[TODO:]
PoseConv3D [22]	Skel.+RGB	[TODO:]	[TODO:]	[TODO:]

5 Experiments

[TODO: 整节数字待 P3 实验产出 (G13). 表格骨架已按分工排好.]

Cross-domain gap. **[TODO: NTU 预训练 zero-shot/finetune 表 (G13)]**

Viewpoint analysis. Per-camera accuracy correlated with the pitch-angle and occlusion statistics of §3.5. **[TODO: 逐相机准确率 + 遮挡消融 (G13)]**

Per-class analysis. Confusion between general and VR-interaction classes; which interaction actions are separable from skeleton alone versus requiring RGB (prop visibility). **[TODO: 混淆矩阵与讨论 (G13)]**

Does ground-truth quality matter? Training the strongest skeleton baseline on raw MoSh++ output versus verified skeletons isolates the value of manual correction (**[TODO: 原始 vs 修正 GT 消融,**

训练两次即可,低成本高说服力 (G23))—an ablation that also serves practitioners deciding whether correction effort is worthwhile.

Zero-shot recognition. [TODO: ActionCLIP 零样本结果 (G13)]

Latency. [TODO: accuracy-latency 曲线与讨论 (G13, G15)]

Qualitative results. [TODO: 成功/失败案例可视化 (G16)]

6 Limitations and Ethics

CAVEAT is captured in a single CAVE installation with 15 actors; the cohort size follows from mocap-suit logistics and is mitigated by the multi-partition X-Sub protocol, but broader demographic coverage is future work. Hand joints are excluded from the official set because sparse hand markers make them unreliable. Training-split skeletons are corrected worst-first rather than exhaustively; per-frame edit masks make the verification status of every joint transparent. All participants gave written informed consent covering public release of RGB video and derived data under an institution-approved protocol ([TODO: 伦理批号, 投稿系统需要 (G17)]); raw uncorrected captures are not distributed. [TODO: 面部是否模糊处理的最终决定 (G17)]

7 Conclusion

CAVEAT turns an operational CAVE into a benchmark, pairing mocap-grade skeletons with the viewpoints, backgrounds, and latency constraints that immersive deployment actually imposes. Our baselines show [TODO: 一句话结论 (G13)], and the released toolchain lowers the cost of extending the corpus. We hope CAVEAT seeds a line of HAR research that treats immersive environments as a first-class setting rather than an afterthought.

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